

ATMOSPHERE – ATMOSPHERIC POLLUTANTS

KEY VOCABULARY

Carbon monoxide: toxic gas from incomplete combustion (CO).

Sulfur dioxide: from sulfur impurities — causes acid rain.

Oxides of nitrogen: form in hot engines — acid rain and smog.

Particulates: soot (carbon) particles — cause health problems.

COMMON MISCONCEPTIONS

~~Carbon dioxide causes acid rain.~~

✓ Acid rain is caused by sulfur dioxide and NO_x.

~~Carbon monoxide has a strong smell.~~

✓ CO is colourless and odourless — hard to detect.

~~Particulates are a gas.~~

✓ Particulates are tiny solid particles (soot).

POLLUTANTS & PROBLEMS

CO → toxic, binds to blood (incomplete combustion).

SO₂ → acid rain (sulfur impurities burn).

NO_x → acid rain & smog (hot engines).

Particulates → global dimming & lung damage.

CO₂ & H₂O → from complete combustion.

1 CARBON MONOXIDE

Explain why carbon monoxide is dangerous and how it forms. [2 marks]

2 ACID RAIN

Name the two gases mainly responsible for acid rain. [2 marks]

3 SULFUR DIOXIDE

Explain where sulfur dioxide comes from when fuels burn. [2 marks]

4 PARTICULATES

State one problem caused by particulates. [1 mark]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Carbon monoxide forms in ____ combustion; ____ dioxide comes from impurities.

incomplete

sulfur

complete

nitrogen

6 NITROGEN OXIDES

Explain how oxides of nitrogen form in a car engine. [2 marks]
